

Topology

1. Sets, relations, functions, and order

1.1. Logic

Theorem 1.1.1 (De Morgan's Laws for statements)

AND and OR interchange under complementation.

In C notation:

$$\begin{aligned}!(A \mid\mid B) &== (!A) \&\& (!B) \\!(A \&\& B) &== (!A) \mid\mid (!B)\end{aligned}$$

In mathematical logical notation:

$$\begin{aligned}\neg(A \vee B) &= (\neg A) \wedge (\neg B) \\ \neg(A \wedge B) &= (\neg A) \vee (\neg B)\end{aligned}$$

Theorem 1.1.2 (De Morgan's Laws for set theory)

Union and intersection interchange under complementation.

$$\begin{aligned}(A \cup B)^c &= A^c \cap B^c \\ (A \cap B)^c &= A^c \cup B^c\end{aligned}$$

Theorem 1.1.3

Given the true statement $P \implies Q$,

- $\neg P \implies \neg Q$ is the **inverse**, which might be true or false.
- $Q \implies P$ is the **converse**, which is true iff the inverse is true.
- $\neg(P \implies Q)$ is the **negation**, which is always false.

- $\neg Q \implies \neg P$ is the **contrapositive**, which is always true.

1.2. Sets

See 01 Set Theory

A set is defined by selecting elements from another set such that $S(x)$ is true:

$$\{x \in A : S(x)\} \text{ or } \{x \in A \mid S(x)\}$$

$\emptyset$ denotes the empty set.

$P \implies Q$ means P implies Q (if P , then Q).

$P \Leftrightarrow Q$ means P and Q are equivalent (P iff Q).

$A \subset B$ means A is a subset of B ($A = B$ is **allowed**).

$A \subsetneq B$ means A is a strict/proper subset of B .

$A = B$ means $A \subset B$ and $B \supset A$.

$A \cup B$ means the union of A and B .

$A \cap B$ means the intersection of A and B .

$A \parallel B$ means $A \cap B = \emptyset$.

$A \setminus B = \{x \in A : x \notin B\}$, everything in A that's not in B .

$A \times B = \{(a, b) : a \in A, b \in B\}$ is called the **Cartesian product** of A and B .

Definition 1.2.1 (Power set)

The set of all subsets of A , denoted $\mathcal{P}(A)$.

1.3. Relations

See 02 Relations

Definition 1.3.1 (Relation)

A **relation** $R : A \rightarrow B$ is any subset of $A \times B$.

aRb means $(a, b) \in R$.

Theorem 1.3.2 (Relations can compose)

If $R : A \rightarrow B$ and $S : B \rightarrow C$, then $S \circ R : A \rightarrow C$ is a subset of $A \times C$.

Warning

$R \circ R^{-1}$ is not necessarily the identity map!

Theorem 1.3.3

$$\begin{aligned}(S \circ R)^{-1} &= R^{-1} \circ S^{-1} \\ (T \circ S) \circ R &= T \circ (S \circ R)\end{aligned}$$

Definition 1.3.4

R is **reflexive** iff aRa for all $a \in A$.

R is **symmetric** iff $aRb \Leftrightarrow bRa$.

R is **transitive** iff $aRb, bRc \Rightarrow aRc$.

1.3.1. Equivalence relations

Definition 1.3.5 (Equivalence relation)

A relation that is reflexive, symmetric, and transitive.

Definition 1.3.6 (Equivalence class)

Let $R : A \rightarrow B$ be an equivalence relation. Then the **equivalence class** of $a \in A$ is

$$[a] = [a]_R := \{b \in A : aRb\}$$

The set of all equivalence relations of R in A is called A/R .

Lemma 1.3.7

Two equivalence classes are either identical or disjoint.

Definition 1.3.8 (Partition)

A collection of disjoint nonempty subsets of A whose union is all of A .

Theorem 1.3.9

Given an equivalence relation on nonempty A , its set of equivalence classes forms a partition of A .

Given a partition P of A , there is a unique equivalence relation whose set of equivalence classes forms P .

1.3.2. Functions (maps)

See 03 Functions

Definition 1.3.10 (Function)

A **function** (also called a **map** or **mapping**) is a relation $f : A \rightarrow B$ where each $a \in A$ appears as the first element (input) of an ordered pair $(a, b) = afb$ exactly once.

If $(a, b) \in f$, we can also say $f(a) = b$. b is the **value** of f at a , or the **image** of a under f .

A is the **domain** and B the **codomain** of f .

Definition 1.3.11 (Injective)

A function f is **injective** or **one-to-one** iff:

- $f(a) = f(b) \Rightarrow a = b$, or equivalently
- $a \neq b \Rightarrow f(a) \neq f(b)$

Definition 1.3.12 (Surjective)

A function $f : A \rightarrow B$ is **surjective** or **onto** iff:

- $f(A) = B$ (its image is its codomain), or equivalently
- for every $b \in B$ there exists $a \in A$ s.t. $f(a) = b$

Definition 1.3.13 (Bijective)

A function is **bijective** and has **one-to-one correspondence** if it is both injective and surjective.

Theorem 1.3.14

A composition of two functions is a function. If both are sur/in/bijective, so is their composition.

Definition 1.3.15

The set of all functions $A \rightarrow B$ is denoted B^A .

Lemma 1.3.16

If A and B are finite, then the size of B^A is $|B|^{|A|}$, the size of B to the size of A .

Theorem 1.3.17

For any set A , there is a bijection between $\mathcal{P}(A)$ and 2^A .

Definition 1.3.18 (Induced map)

The **induced map** of the relation $f : A \rightarrow B$ is a function $\bar{f} : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ defined as

$$\bar{f}(X) := \{b \in B : b = f(a) \text{ for some } x \in A\}$$

$\bar{f}(X)$ is called the **image** of X . This is sometimes denoted $f(X)$ (which is an abuse of notation).

$\bar{f}$ is always a function regardless of if f is a function.

1.4. Ordering

1.4.1. Partial and total ordering

See 04 Ordering

Definition 1.4.1 (Partial order)

A **partial order** $\leq$ on a set A is a relation that is

- **reflexive**: $a \leq a$ for all $a \in A$
- **transitive**: $a \leq b$ and $b \leq c \Rightarrow a \leq c$
- **antisymmetric**: $a \leq b$ and $b \leq a \Rightarrow a = b$

Given a partial order $\leq$, we define the partial order $\geq$ so that $a \geq b \Leftrightarrow b \leq a$.

Definition 1.4.2 (Total order)

A **total order** $\leq$ on a set A is a partial order where for all $a, b \in A$, $a \leq b$, $a \geq b$, or both.

Definition 1.4.3 (Strict order)

A **strict order** $<$ determined by a partial order $\leq$ is defined s.t. $a < b$ iff $a \leq b$ and $a \neq b$.

Definition 1.4.4 (Partially ordered and totally ordered set)

A **partially ordered set** or **poset** is a pair of a set and a partial order for that set.

A **totally ordered set** is a pair of a set and a total order for that set.

By abuse of notation we also say that the set itself is partially/totally ordered.

Definition 1.4.5 (Upper and lower bounds)

Given a poset A and subset $S \subset A$:

An **upper bound** for S is an element $a \in A$ s.t. $s \leq a$ for all $s \in S$.

A **lower bound** for S is an element $a \in A$ s.t. $a \leq s$ for all $s \in S$.

The **supremum** of S , $\sup S$, is the upper bound such that for any other upper bound u of S , $\sup S \leq u$.

The **infimum** of S , $\inf S$, is the lower bound such that for any other lower bound l of S , $l \leq \inf S$.

A set does not have to have a supremum or infimum, but if it does it is unique.

1.4.2. Lattices

See 04 Ordering.2

Definition 1.4.6 (Lattice)

A **lattice** is a poset in which every two-element subset has both a supremum and infimum.

A **complete lattice** is a poset in which every (including infinite) subset has both a supremum and infimum.

Theorem 1.4.7

The power set of any set, ordered by inclusion, is a complete lattice.

1.4.3. Order-preserving

Definition 1.4.8 (Order-preserving map)

A function $f : A \rightarrow B$ between posets is **order-preserving** iff whenever $a_1 \leq a_2 \implies f(a_1) \leq f(a_2)$.

Theorem 1.4.9

Every order-preserving map of a complete lattice to itself must have a fixed point (an element of the domain for which $f(a) = a$).

Corollary 1.4.9.1

Let $f : A \rightarrow B$ and $g : B \rightarrow A$. Then, A and B can be split such that:

- A is split into A_1 and A_2 such that $A_1 \cup A_2 = A$ and $A_1 \cap A_2 = \emptyset$
- B is split A_1 and A_2 such that $A_1 \cup A_2 = A$ and $A_1 \cap A_2 = \emptyset$
- $\bar{f}(A_1) = B_1$ and $\bar{g}(B_2) = A_2$

Definition 1.4.10 (Order isomorphism)

An order-preserving bijection.

1.4.4. Well-ordering

Definition 1.4.11 (Well-ordered set)

A **well-ordered set** is a totally-ordered set in which every nonempty subset has a least element.

Equivalently, a set where every element has a “next element” greater than it.

Definition 1.4.12 (Initial segment)

Given poset A , with $a \in A$, the **initial segment** of a is the set $\{a' \in A : a' < a\}$.

Lemma 1.4.13

Let f be an order-preserving injection of a well-ordered set into itself. Then $a \leq f(a)$ for all $a \in A$.

A well-ordered set is not order isomorphic to any of its initial segments.

Theorem 1.4.14

Let A or B be well-ordered sets. Then exactly one of the following is true:

1. A is order isomorphic to B .
2. A is order isomorphic to an initial segment of B .
3. B is order isomorphic to an initial segment of A .

1.5. Cardinality

Assume that there is some “overset” containing all of the sets we consider.

Definition 1.5.1 (Cardinality)

$\text{card}(A)$ is the equivalence class of A under bijection

Equivalently, $\text{card}(A) = \text{card}(B)$ iff there is a bijection $A \rightarrow B$.

Two sets are **equipotent** if they have the same cardinality.

Definition 1.5.2

$\text{card } A \leq \text{card } B$ iff there is an injection from A to B .

Theorem 1.5.3 (Schroeder-Bernstein Theorem)

$$\text{card}(A) \leq \text{card } B \text{ AND } \text{card}(B) \leq \text{card}(A) \\ \implies \text{card}(A) = \text{card}(B)$$

Theorem 1.5.4 (Cantor’s Theorem)

For any set A ,

$$\text{card}(A) < \text{card}(\mathcal{P}(A))$$

1.6. Countability

Definition 1.6.1 (Countability)

S is countable iff there is an injection $f : S \rightarrow \mathbb{Z}^+$, or equivalently iff $\text{card } S \leq \text{card } \mathbb{Z}^+$.

Corollary 1.6.1.1

1. Every subset of a countable set is countable.
 - The elements of a countable set can be indexed by the positive integers.
 - All finite sets are countable.
2. A countable union of countable sets is countable.

Theorem 1.6.2

The product of a finite number of countable sets is countable.

However, a product of an infinite number of countable sets is not countable.

Theorem 1.6.3

$$\text{card}(\mathbb{R}) = \text{card}(\mathcal{P}(\mathbb{Z}^+))$$

Thus, the reals are not countable.

1.7. Axiom of choice

Axiom 1.7.1 (Axiom of Choice)

Given a infinite collection of disjoint finite sets, it is possible to select one element from each of the sets.

Equivalently, if A is a set of nonempty sets, there is always a function $f : A \rightarrow \cup A$ such that $f(a) \in a$ for all $a \in A$.

In this class, we assume the Axiom of Choice.

Theorem 1.7.2 (Zermelo's Theorem)

The following theorems are equivalent:

- the Axiom of Choice
- the Well-Ordering Property
- the Hausdorff Maximum Principle
- Zorn's Lemma

Theorem 1.7.3 (Well-Ordering Property (Zermelo's Postulate))

Every set can be well-ordered.

Theorem 1.7.4 (Hausdorff Maximum Principle)

Every poset has a "maximal" totally ordered subset, i.e. a totally ordered subset that cannot be extended without escaping the poset.

Theorem 1.7.5 (Zorn's Lemma)

If A is a poset in which every totally ordered subset has an upper bound, then A has a maximal element, i.e. there is no larger element in the poset.

2. Topologies

2.1. Metric spaces

See 06 Metric Spaces

Definition 2.1.1 (Metric)

Given a set M , a **metric** $d : M \times M \rightarrow \mathbb{R}$ on M is a positive definite symmetric function satisfying the triangle inequality, i.e. it satisfies:

- **Positivity:** $d(x, y) \geq 0$ for all $x, y \in M$
- **Definiteness:** $d(x, y) = 0$ iff $x = y$
- **Symmetry:** $d(x, y) = d(y, x)$
- **Triangle Inequality:** $d(x, y) + d(y, z) \geq d(x, z)$ for all $x, y, z \in M$

Such a function satisfies everything we expect from a distance.

A **pseudometric** satisfies all the above properties except definiteness.

A set equipped with a (pseudo)metric is a **(pseudo)metric space**.

Theorem 2.1.2 (Metric characterization of continuity)

$f : M \rightarrow N$ is continuous at $x \in M$ iff for any $\varepsilon > 0$, there exists $\delta > 0$ s.t. $d(x, y) < \delta \implies d(f(x), f(y)) < \varepsilon$.

2.1.1. Epsilon neighborhoods

Definition 2.1.3 (Epsilon neighborhood)

Let N be a metric space.

An **epsilon neighborhood** of a set $A \subset N$ is the set

$$U(A, \varepsilon) := \{y \in N : (\exists x \in A : d(x, y) < \varepsilon)\}$$

Equivalently, the set of all points within distance ε of some point in A .

An **epsilon** or **open neighborhood**, **disk**, or **ball** of radius ε around a point x is defined as the set of all points within distance ε of x :

$$U(x, \varepsilon) := \{y \in N : d(x, y) < \varepsilon\} = U(\{x\}, \varepsilon)$$

Theorem 2.1.4 (Epsilon neighborhood definition of continuity)

$f : M \rightarrow N$ is continuous at $x \in M$ iff for every open set $V \subset N$ that contains $f(x)$, there is an open set $U \subset M$ containing x s.t. $f(U) \subset V$.

2.1.2. Subspaces

Definition 2.1.5 (Subspace of a metric space)

If $A \subset M$ where d is a metric on M , then $d|_A := d$ restricted to A is a metric on A .

A with this metric is a **subspace** of M .

2.1.3. Common metrics

Definition 2.1.6 (Standard metric on $\mathbb{R}^n$)

This is the typical distance on $\mathbb{R}^n$.

$$\mu_2(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Definition 2.1.7 (Taxicab metric on $\mathbb{R}^n$)

Also called the **Manhattan distance**. It's the distance if you can only travel parallel to one of the axes.

$$\mu_1(x, y) := \sum_{i=1}^n |x_i - y_i|$$

Definition 2.1.8 (Chebyshev metric on $\mathbb{R}^n$)

The greatest single difference between any pair of corresponding coordinates of the two points:

$$\mu_\infty(x, y) := \max |x_i - y_i|$$

Definition 2.1.9 (Paris metro metric)

The distance if you can only travel to and from the origin, and all other travel must go through the origin.

$$\mu_{\text{pm}}(x, y) := d(x, 0) + d(0, y)$$

Definition 2.1.10 (Discrete metric)

The distance between any two distinct points is 1.

$$\mu_0(x, y) := \begin{cases} 1 & \text{for } x \neq y \\ 0 & \text{for } x = y \end{cases}$$

Definition 2.1.11 (Indiscrete pseudometric)

The distance between everything is 0:

$$\mu_{\text{triv}}(x, y) := 0$$

Also called the *trivial pseudometric*.

2.2. Topologies

See 07 Topologies

Definition 2.2.1 (Topology)

A **topology** $\mathcal{T}$ on the set X is a collection of subsets of X , called the **open sets**, such that

1. $\mathcal{T}$ is closed under arbitrary unions.
2. $\mathcal{T}$ is closed under finite intersections.

A set together with a topology on it is called a **topological space**.

Definition 2.2.2 (Closed set)

The complement of an open set.

Lemma 2.2.3

The empty set and whole space are always both closed and open.

Definition 2.2.4 (Metrizable topology)

A topology that is the set of open sets given by some metric.

Definition 2.2.5 (Finer, coarser topologies)

A topology is **coarser** (also called **smaller** or **weaker**) than another topology iff it is a subset of the other.

A topology is **finer** (also called **larger** or **stronger**) iff it is a superset of the other.

Lemma 2.2.6

The set of all topologies on a specific set is a lattice.

2.2.1. Common topologies

Definition 2.2.7 (Standard topology on $\mathbb{R}$)

A set is open iff it is the union of open intervals.

Definition 2.2.8 (Discrete topology)

All sets are open:

$$\mathcal{T} = \mathcal{P}(X)$$

This is equivalent to the topology given by the discrete metric.

Definition 2.2.9 (Indiscrete topology)

Only the entire topological space and $\emptyset$ are open.

This is a metrizable space given by the indiscrete pseudometric.

Definition 2.2.10 (Metric topology)

The topology generated by the base consisting of all open balls $U(x, \varepsilon)$.

Definition 2.2.11 (Cofinite topology)

All subsets whose complements are finite are open. $\emptyset$ is also open.

Definition 2.2.12 (Sorgenfrey line)

The topology generated by the base consisting of all half-open intervals $[a, b)$.

2.3. Closure, interior, boundary

See 07 Topologies.2

Definition 2.3.1 (Closure)

Given $A \subset X$ where X is a topological space, the **closure** of A , denoted $\overline{A}$ or $\text{cl } A$, is the intersection of all closed subsets containing A .

Theorem 2.3.2 (Kuratowski closure axioms)

The closure operation satisfies

1. $A \subset \overline{A}$
2. $\overline{\overline{A}} = \overline{A}$
3. $\overline{A \cup B} = \overline{A} \cup \overline{B}$
4. $\overline{\emptyset} = \emptyset$

Definition 2.3.3 (Interior)

Given $A \subset X$ where X is a topological space, the **interior** of A , A° , is the union of all open sets contained in A .

Definition 2.3.4 (Boundary)

Given $A \subset X$ where X is a topological space, the **boundary** or **frontier** of A , denoted ∂A or $\text{bd } A$, is

$$\partial A := \overline{A} - A^\circ$$

Some authors use the word “boundary” to mean only the part of the frontier contained in A .

Definition 2.3.5 (Exterior)

Given $A \subset X$ where X is a topological space, the **exterior** is the complement of the closure of A :

$$X - \overline{A} = \overline{A}^c$$

Theorem 2.3.6

Given $A \subset X$ where X is a topological space, X is partitioned into three disjoint parts:

$$X = A^\circ \sqcup \partial A \sqcup X - \overline{A}$$

2.3.1. Bases

See 08 Neighborhoods and Bases.3

Definition 2.3.7 (Base)

Let $\mathcal{T}$ be a topology on X . A **base** for $\mathcal{T}$ is a collection of open sets such that any other open set can be made out of unions of the elements of the base.

Definition 2.3.8 (Second countable)

A space is **second countable** iff it has a countable base.

Lemma 2.3.9

All second countable spaces are first countable.

Example. Let $\mathcal{T}$ be a countable discrete space. Then the base is at least all the singletons, which is

uncountably many. Thus, $\mathcal{T}$ is not second countable.

Definition 2.3.10 (Sub-base)

A collection of sets such that finite intersections of the sets in the sub-base form a base.

Definition 2.3.11 (Topology generated by a sub-base)

The topology generated by a sub-base B is defined such that a set is open iff:

- it is a member of B , or
- it is a finite intersection of members of B , or
- it is an arbitrary union of those intersections

Theorem 2.3.12 (Topology generated by a base)

The topology generated by a base B such that a set is open iff it is an arbitrary union of members of B .

Example. The set of infinite intervals with one open side is a sub-base of $\mathbb{R}$ in the standard topology.

2.4. Neighborhoods

See 08 Neighborhoods and Bases

Definition 2.4.1 (Neighborhood)

Given $A \subset X$ (where X is a topological space), N is a **neighborhood** of A iff N contains an open set containing A .

Given $x \in X$, N is a neighborhood of x iff N is a neighborhood of $\{x\}$, i.e. it contains an open set containing x .

$\mathcal{N}(x)$ denotes the set of all neighborhoods of x , which is called the **neighborhood system** or **filter** for x .

Most of the things we can prove about neighborhoods we can also prove about open neighborhoods, so some authors define neighborhoods to always be open.

Lemma 2.4.2

$U \subset X$ is open iff for each $x \in U$ there is a neighborhood N_x of x contained in U .

Axiom 2.4.3 (Hausdorff neighborhood axioms)

Let $x \in X$. Then,

1. x is in every one of its neighborhoods:

$$N \in \mathcal{N}(x) \implies x \in N$$

2. $\mathcal{N}(x)$ is closed under supersets:

$$N \in \mathcal{N}(x), N \subset M \implies M \in \mathcal{N}(x)$$

3. $\mathcal{N}(x)$ is closed under finite intersection.

4. For each neighborhood N of x , it contains a neighborhood N' such that for each item y in N' , N is a neighborhood of y :

$$N \in \mathcal{N}(x) \implies (\exists N' \in \mathcal{N}(x) :$$

$$N' \subset N \text{ and } y \in N' \implies N \in \mathcal{N}(y))$$

Conversely, if each $x \in X$ has a collection of neighborhoods $\mathcal{N}'(x)$ satisfying the above axioms, then there is a unique topology on X for which $\mathcal{N}'(x) = \mathcal{N}(x)$ for all $x \in X$.

Theorem 2.4.4 (Neighborhood characterization of closure)

Let $x \in X$, $A \subset X$. Then $x \in \overline{A}$ iff every neighborhood of x meets A .

Theorem 2.4.5 (Neighborhood characterization of boundary)

∂A is the set of all points $x \in X$ s.t. every neighborhood of x meets both A and $X - A$.

2.4.1. Neighborhood bases

Definition 2.4.6 (Neighborhood base)

Let $x \in X$. A **neighborhood base** of x is a set $\mathcal{B}(x) \subset \mathcal{N}(x)$ such that for any $N \in \mathcal{N}(x)$ there is a $B \in \mathcal{B}(x)$ such that $B \subset N$.

Equivalently, it is a set of **basic neighborhoods** around a x s.t. each neighborhood of x contains a basic neighborhood.

Equivalently, it is a filter base of the neighborhood filter $\mathcal{N}$.

Definition 2.4.7 (First countable)

A space is **first countable** iff if each point has a countable neighborhood base.

Example. Let $\overline{\Omega}$ be the minimal uncountable well-ordered set with its endpoint ω_0 . $\mathcal{B}(\omega_0)$ is not countable, so $\overline{\Omega}$ is not first countable.

Example. Every metrizable space is first countable.

Axiom 2.4.8 (Neighborhood axioms)

Let $\mathcal{B}(x)$ be a neighborhood base at x . Then

1. There exist basic neighborhoods of x .
2. x is contained in each of its basic neighborhoods:

$$U \in \mathcal{B}(x) \implies x \in U$$

3. $\mathcal{B}(x)$ is closed under finite intersection:

$$U_1, U_2 \in \mathcal{B}(x) \implies (\exists V \in \mathcal{B}(x) : V \subset U_1 \cap U_2)$$

4. For each basic neighborhood U of x , it contains a basic neighborhood V such that for

each item y in V , V contains a basic neighborhood W of y :

$$U \in \mathcal{B}(x) \implies (\exists V \in \mathcal{B}(x) :$$

$$V \subset U \text{ and } (y \in V \implies \exists W \in \mathcal{B}(y) : W \subset V))$$

Conversely, given a set $\mathcal{B}(x)$ for each $x \in X$ satisfying the above axioms, there is a unique topology $\mathcal{T}$ on X s.t. for all $x \in X$, $\mathcal{B}(x)$ is a neighborhood base of x in the topology $\mathcal{T}$. $\mathcal{T}$ is defined by:

$$\mathcal{T} := \{U \subset X : x \in U \implies (\exists V \in \mathcal{B}(x) \ni V \subset U)\}$$

2.4.2. Accumulation points

Definition 2.4.9 (Accumulation point)

x is an **accumulation point** of A iff every neighborhood of x contains a point of A other than x .

Definition 2.4.10 (Derived set)

The set of accumulation points of A is called the **derived set**, denoted A' .

Corollary 2.4.10.1 (Accumulation point characterization of closure)

$$\overline{A} = A \cup A'.$$

2.5. Continuity

See 09 Continuous Functions

Definition 2.5.1 (Continuous function)

$f : X \rightarrow Y$ is a continuous function at $x_0 \in X$ iff $f^{-1}(N)$ is a neighborhood of x_0 for every neighborhood N of $f(x_0)$.

Equivalently, it is continuous iff every neighborhood of $f(x)$ pulls back (by f^{-1}) to a neighborhood of x .

f is continuous if it is continuous at all $x_0 \in X$.

Definition 2.5.2 (Homeomorphism)

f is a homeomorphism iff f is a continuous bijection whose inverse is also continuous.

Homeomorphisms are equivalence relations between topological spaces.

Theorem 2.5.3 (Continuity composes)

If $f : X \rightarrow Y$ is continuous at x_0 and $g : Y \rightarrow Z$ is continuous at $f(x_0)$, then $g \circ f : X \rightarrow Z$ is continuous at x_0 .

Theorem 2.5.4

Given $f : X \rightarrow Y$, the following are equivalent.

1. f is continuous.
2. $f(\overline{A}) \subset \overline{f(A)}$ for all $A \subset X$.
3. If $B \subset Y$ is closed, then $f^{-1}(B)$ is closed in X .
4. If $B \subset Y$ is open, then $f^{-1}(B)$ is open in X .

Lemma 2.5.5

$f : X \rightarrow Y$ is continuous iff $f^{-1}(c)$ is open for each c in some subbase of Y .

Lemma 2.5.6

Any function from any discrete space to any other space is continuous.

Any function from any space to any indiscrete space is continuous.

2.6. Initial (induced) topology

See 10 Induced Topologies

2.6.1. Subspace topology

Definition 2.6.1 (Subspace topology)

Given $A \subset X$, where X is a topological space, the **subspace topology (relative topology)** on A is defined such that $B \subset A$ is open in A iff $B = A \cap O$ for some open set O in X .

The subspace topology is the initial topology induced by the inclusion map $i_A : A \rightarrow X$ defined by $i_A(x) := x$.

A equipped with the subspace topology is called a **subspace** of X .

Example. $\mathbb{Q}^\circ$ under the standard topology on $\mathbb{R}$ is empty, since there are no open intervals in $\mathbb{Q}$.

However, $\mathbb{Q}^\circ$ under the subspace topology on $\mathbb{Q}$ is the entire space, since all spaces are open.

Lemma 2.6.2

The following properties are preserved by taking subspaces:

- discrete
- indiscrete
- metrizable
- subspace of some space

2.6.2. Initial (induced) topology

Definition 2.6.3 (Initial topology)

For each α in some index set $\mathcal{A}$, let A_α be a topological space and f_α be a function $X \rightarrow A_\alpha$. Then, the initial topology **induced** on X by the set of f_α is the coarsest topology in X for which each f_α is continuous.

Equivalently, it is the intersection of all of the topologies for which each f_α is continuous.

Also called: **initial topology**, **strong topology**, **limit topology**, **projective topology**

Equivalently, given a bunch of sets A_α and functions $f_\alpha : X \rightarrow A_\alpha$, pull the open sets in each A_α back to X (by f_α), and define these pullbacks to be open on X . This is a sub-base; generate a topology from this sub-base by defining the finite intersections and arbitrary unions to be open.. This forms a topology on X , which is called the **initial topology**.

Lemma 2.6.4

The initial topology on X induced by $\{f_\alpha\}$ is the topology generated by the following sub-base: the collection of all $f_\alpha^{-1}(U)$ for all open sets $U \subset X_\alpha$.

Theorem 2.6.5

The topology induced by $\{f_\alpha\}$ is characterized by if Y is any topological space, then $g : Y \rightarrow X$ is continuous iff each $g \circ f_\alpha : Y \rightarrow A_\alpha$.

Definition 2.6.6 (Open and closed functions)

Let X and Y be topological spaces. $f : X \rightarrow Y$ is **open** iff $f(U)$ is open for all U open on X .

$f : X \rightarrow Y$ is **closed** iff $f(C)$ is closed for all C closed on X .

(These are not opposite.)

2.6.3. Product topology

See Munkres 2§19 (p.113)

Definition 2.6.7 (Projection)

Let $\{X_\alpha\}$ be a collection of sets, and $X = \prod_{\alpha \in A} X_\alpha$. Then for each $\alpha \in A$, the **projection map** $\pi_\alpha : X \rightarrow X_\alpha$ maps each tuple in A to its α th element.

Definition 2.6.8 (Product topology)

Let $\{X_\alpha\}$ be a set of topological spaces, and $X = \prod_{\alpha \in A} X_\alpha$. The topology on X induced by the collection of projections $\{\pi_\alpha : \alpha \in A\}$ is called the **product topology**.

Equivalently, given a product space X , pull the open sets in each X_α back to X (using the projection maps π_α), and define these pullbacks to be open on X . This is a sub-base; generate a topology from this sub-base by defining the finite intersections and arbitrary unions to be open. This forms a topology on X , which is called the **product topology**.

Lemma 2.6.9 (Product topology constrains finitely many axes)

The collection of sets which are products of open sets in each of the X_α , where all but a finite number are the entire set X_α , form a base for the product topology.

Thus, only a finite number of axes can be constrained when creating an open set in the product topology.

This is because the pullbacks of open sets in one of the X_α only constrain a single axis, and we can only take finite intersections of them.

Theorem 2.6.10

If $\mathcal{A}' \subset \mathcal{A}$, then $\pi : \prod_{\alpha \in \mathcal{A}} X_\alpha \rightarrow \prod_{\alpha \in \mathcal{A}'} X_\alpha$ is continuous and open.

2.6.4. Box topology

Definition 2.6.11 (Box topology)

Let $\{X_\alpha\}$ be a set of topological spaces, and $X = \prod_{\alpha \in A} X_\alpha$. The **box topology** on X is generated by the base

$$\left\{ \prod_{\alpha \in A} U_{\alpha} : U_{\alpha} \text{ open in } X_{\alpha} \right\}$$

2.7. Covers and defining families

See 10 Induced Topologies.3

Definition 2.7.1 (Defining family)

A **defining family** for topological space X is a collection $\mathcal{C}$ s.t. $U \subset X$ is open in X iff its intersection with each $C \in \mathcal{C}$ is open in the subspace topology on C .

Definition 2.7.2 (Open cover)

An **open cover** of X is a collection of open sets whose union contains all of X .

All open covers are defining families.

Definition 2.7.3 (Locally finite)

A collection of subsets of a topological space is **locally finite** if each point has a neighborhood that meets only finitely many sets in the collection.

Definition 2.7.4 (Closed cover)

An **open cover** of X is a collection of closed sets whose union contains all of X .

Locally finite closed covers are defining families.

Theorem 2.7.5

Let $\mathcal{C}$ be a defining family for X . Then $f : X \rightarrow Y$ is continuous iff its restriction to each $C \in \mathcal{C}$ is continuous.

2.8. Final (co-induced) topologies

Definition 2.8.1 (Final topology)

For each α in some index set $\mathcal{A}$, let A_{α} be a topological space and g_{α} be a function $A_{\alpha} \rightarrow Y$. Then, the final topology **coinduced** on Y by the set of g_{α} is the finest topology in Y for which each g_{α} is continuous.

Equivalently, it is the union of all of the topologies for which each g_{α} is continuous.

Also called: **final topology**, **weak topology**, **colimit topology**, **inductive topology**

Equivalently, a set U is open in the final topology iff its preimage $g_{\alpha}^{-1}(U)$ is open in the topology on Y .

2.8.1. Quotient topology

Definition 2.8.2 (Quotient topology)

Given topological space X and surjection $g : X \rightarrow Y$, the final topology coinduced by g is the **quotient topology**.

Then, Y is the **quotient space** of X by g .

3. Limits

3.1. Sequences

Definition 3.1.1 (Sequence)

A function $x : \mathbb{N} \rightarrow X$, where X is a topological space. This is denoted $\{x_n\}_{n=1}^{\infty}$, where $x_n = x(n) \in X$.

Definition 3.1.2 (Convergence of a sequence)

x_n converges to x_0 ($x_n \rightarrow x_0$) iff for every neighborhood U around x_0 , there exists N_U s.t. $n > N_U$ implies $x_n \in U$. Equivalently, every

neighborhood x_0 contains all but finitely many of the x_n .

Lemma 3.1.3

In a metric space, $x_n \rightarrow x_0$ iff

$$\lim_{n \rightarrow \infty} d(x_n, x_0) = 0$$

Lemma 3.1.4

In the product space $X = \prod X_{\alpha}$, $x_n \rightarrow x_0$ in X iff $\pi_{\alpha}(x_n) \rightarrow \pi_{\alpha}(x_0)$ for every α .

Theorem 3.1.5

In a first countable space, $x \in \overline{A}$ iff there is a sequence of points in A that converge to x .

Theorem 3.1.6

If $f : X \rightarrow Y$ is continuous at x_0 and $\{x_n\} \rightarrow x_0$, then $f(x_n) \rightarrow f(x_0)$.

Conversely, if x_0 has a countable neighborhood base and each sequence $\{x_n\} \rightarrow x_0$, $f(x_n) \rightarrow f(x_0)$, then f_0 is continuous at x_0 .

3.2. Nets

Definition 3.2.1 (Directed upward)

A poset in which any two elements have an upper bound. That is, given $\alpha, \beta \in S$, there exists $\gamma \in S$ s.t. $\alpha \leq \gamma$ and $\beta \leq \gamma$

Definition 3.2.2 (Net)

A subset of a topological space which is directed upward by some $\leq$ (which is not necessarily related to the space.)

This is sometimes written using a directed-upward set $\mathcal{A}$, where the net is denoted $\{x_{\alpha}\}_{\alpha \in \mathcal{A}}$

Definition 3.2.3 (Convergence of a net)

$x_\alpha \rightarrow x_0$, where $\{x_\alpha\}_{\alpha \in \mathcal{A}}$ is a net, iff for each neighborhood N around x_0 , there exists $\gamma \in \mathcal{A}$ s.t. $x_\alpha \in N$ for all $\alpha \geq \gamma$.

Lemma 3.2.4 (Net characterization of closure)

If A is a subset of a topological space, $x_0 \in \overline{A}$ iff there exists a net in A converging to x_0 .

Lemma 3.2.5 (Net characterization of continuity)

$f : X \rightarrow Y$ is continuous at x_0 iff for each net $x_\alpha \rightarrow x_0$, $f(x_\alpha) \rightarrow f(x_0)$.

3.3. Filters

Definition 3.3.1 (Filter base)

A **filter base** $\mathcal{F}$ in the set X is a nonempty collection of nonempty subsets of X which is directed downward by inclusion.

Definition 3.3.2 (Filter)

A filter base $\mathcal{F}$ which is closed under supersets:

$$\forall A \in \mathcal{F} \quad \nexists B \in X : A \subsetneq B$$

Lemma 3.3.3

The collection of all sets in a filter base together with their supersets is a filter.

Definition 3.3.4 (Canonical filter)

The **canonical filter generated by A** is the set of all supersets of A , if A is nonempty.

Definition 3.3.5 (Fréchet filter)

The collection of subsets of the nonnegative integers whose complements are finite.

Lemma 3.3.6 (Images of filter bases are filter bases)

If $\mathcal{F}$ is a filter base on X , and $f : X \rightarrow Y$, then

$$f(\mathcal{F}) = \{f(F) : F \in \mathcal{F}\}$$

is a filter base on Y .

Definition 3.3.7 (Elementary filter)

The image of the Fréchet filter.

Definition 3.3.8 (Convergence of a filter base)

$\mathcal{F} \rightarrow x_0$ iff every neighborhood N of x_0 contains some $F \in \mathcal{F}$.

Lemma 3.3.9 (Constructing filter base from net)

Given the nonempty net $\{x_\alpha\}_{\alpha \in \mathcal{A}}$ in X , we can define

$$X_\alpha = \{x_{\alpha'} : \alpha' < \alpha\}$$

Then, $\{X_\alpha\}$ is a filter base in X , and $x_\alpha \rightarrow x_0$ iff $X_\alpha \rightarrow x_0$.

Lemma 3.3.10 (Collection of neighborhoods is filter)

The collection of all neighborhoods of any nonempty set is a filter.

Lemma 3.3.11 (Filter characterization of continuity)

Let $f : X \rightarrow Y$, and $x_0 \in X$ where N is the set of neighborhoods of x_0 . Then, f is continuous at x_0 iff $N \rightarrow f(x_0)$.

Lemma 3.3.12 (Filter base characterization of closure)

$x \in \overline{A}$ iff there is a filter base in A converging to x .

Theorem 3.3.13

Given a filter base $\mathcal{F}$ in the product space $X = \prod X_\alpha$, $\mathcal{F} \rightarrow x_0$ iff each $\pi_{\alpha(\mathcal{F})} \rightarrow \pi_\alpha(x_0)$ in X_α .

3.3.1. Ultrafilters

Lemma 3.3.14

The set of all filter bases on a specific set is a poset under inclusion.

The set of all filter bases containing a given filter base is a poset where every totally ordered subset has an upper bound.

Definition 3.3.15 (Ultrafilter)

The maximal filter base in a specific set; i.e. the filter base to which no more sets can be added while maintaining downward direction.

The ultrafilter is a filter.

Theorem 3.3.16 (Test for ultrafilter)

A filter base $\mathcal{F}$ is an ultrafilter iff for each $E \subset X$ either $E \in \mathcal{F}$ or $X - E \in \mathcal{F}$.

Definition 3.3.17 (Cluster point)

A **cluster point** of a filter base $\mathcal{F}$ is a point that is in $\overline{F}$ for every $F \in \mathcal{F}$.

Theorem 3.3.18

x is a cluster point of $\mathcal{F}$ iff there exists an ultrafilter $\mathcal{F}'$ containing $\mathcal{F}$ that converges to x .

4. Separation

4.1. Separation properties

Definition 4.1.1 (Separable by neighborhoods)

A and B in the space X are **separable by neighborhoods** iff there exist neighborhoods U of A and V of B s.t. $U \cap V = \emptyset$.

This is equivalent to being **separable by open sets**.

Definition 4.1.2 (Separation by closed neighborhoods)

A and B in the space X are **separable by closed neighborhoods** iff there exist neighborhoods U of A and V of B s.t. $\overline{U} \cap \overline{V} = \emptyset$.

Definition 4.1.3 (Separation by continuous functions)

A and B in the space X are **separable by continuous functions** iff there exists a continuous function such that $f(x) = 0$ for all $x \in A$ and $f(x) = 1$ for all $x \in B$.

4.2. T_0 through T_2 spaces

Definition 4.2.1 (Kolmogorov space (T_0))

A space where given two distinct points in the space, at least one has a neighborhood that does not include the other.

Lemma 4.2.2

X is T_0 iff for any $x, y \in X$, $x \neq y \Leftrightarrow \overline{\{x\}} \neq \overline{\{y\}}$.

Definition 4.2.3 (Fréchet space (T_1))

A space where given two distinct points in the space, each has a neighborhood that does not include the other.

T_1 spaces are T_0 .

Lemma 4.2.4

Points in T_1 spaces are closed sets.

Lemma 4.2.5

X is T_1 iff $\{x\} = \overline{\{x\}}$ for all $x \in X$.

Definition 4.2.6 (Hausdorff space (T_2))

A space where given two distinct points $x \neq y$, there are neighborhoods X of x and Y of y s.t. $X \cap Y = \emptyset$.

Equivalently, a space where all two points are separable by neighborhoods.

T_2 spaces are T_1 .

Definition 4.2.7 ($T_{2\frac{1}{2}}$ space)

A space where given two distinct points $x \neq y$, there are neighborhoods X of x and Y of y s.t. $\overline{U} \cap \overline{V} = \emptyset$.

Equivalently, a space where all two points are separable by closed neighborhoods.

This is also called **completely Hausdorff** (in older texts) and **Urysohn** (in newer texts).

$T_{2\frac{1}{2}}$ spaces are T_2 .

Definition 4.2.8 (Functionally Hausdorff space)

A space X where given two points $x \neq y$, there is a continuous function $f : X \rightarrow [0, 1]$ with $f(x) = 0$ and $f(y) = 1$.

Equivalently, a space where all two points are separable by continuous functions.

This is also called **Urysohn** (in older texts) and **completely Hausdorff** (in newer texts).

Functionally Hausdorff spaces are $T_{2\frac{1}{2}}$.

Lemma 4.2.9

The following are equivalent:

1. X is T_2
2. Limits of filter bases in X are unique.
3. The diagonal, $\Delta = \{(x, y) \in X \times X : x = y\}$

Lemma 4.2.10

Products and subspaces of T_i spaces are T_i for $i = 0, 1, 2$.

4.3. Regular spaces

Definition 4.3.1 (Regular space)

A space in which any point and any disjoint closed set can be separated by neighborhoods.

Equivalently a space in which any point and any disjoint closed set can be separated by closed neighborhood.

Theorem 4.3.2

X is regular iff the closed neighborhoods of each $x \in X$ form a neighborhood base of x .

Definition 4.3.3 (Regular Hausdorff space (T_3))

A T_1 space which is regular.

Equivalently, a $T_{2\frac{1}{2}}$ space which is regular.

Definition 4.3.4 (Completely regular space)

A space in which closed sets and disjoint single points can be separated by continuous functions.

Definition 4.3.5 (Tychonoff space ($T_{3\frac{1}{2}}$))

A completely regular T_1 space.

Theorem 4.3.6

Subspaces and products of regular spaces are regular.

Subspaces and products of completely regular spaces are completely regular.

Theorem 4.3.7

X is completely regular iff it has the topology induced by its set of bounded continuous real-valued functions.

Corollary 4.3.7.1

A topological space is Tychonoff iff it is homeomorphic to a subspace of a hypercube $([0, 1]^n)$.

4.4. Normal spaces

Definition 4.4.1 (Normal space)

A topological space is **normal** iff disjoint closed sets are separable by neighborhoods.

Definition 4.4.2 (Normal Tychonoff space (T_4))

A normal T_1 space.

Equivalently, a normal $T_{3\frac{1}{2}}$ space.

Lemma 4.4.3

If X contains a dense set D and a closed discrete subspace S with cardinality at least as large as $\mathcal{P}(D)$, then X cannot be normal.

Definition 4.4.4 (Completely normal)

A space in which every subspace is normal.

Definition 4.4.5 (Completely normal Tychonoff (T_5))

A completely normal T_1 space.

Equivalently, a completely normal T_4 space.

Theorem 4.4.6 (Urysohn's Lemma)

In a normal space X , any two disjoint closed sets A and B can be separated by a continuous function.

Theorem 4.4.7 (Tietze Extension Theorem)

X is normal iff for any closed set A of X , every continuous function $f : A \rightarrow [0, 1]$ can be extended to a continuous function $X \rightarrow \mathbb{R}$.

Corollary 4.4.7.1

X is normal iff for any closed set A of X , every continuous function $f : A \rightarrow \mathbb{R}$ can be extended to a continuous function $X \rightarrow \mathbb{R}$.

5. Countability properties

See 17 Countability Properties

Definition 2.4.7 (First countable)

A space is **first countable** iff if each point has a countable neighborhood base.

Definition 2.3.8 (Second countable)

A space is **second countable** iff it has a countable base.

Definition 5.1 (Separable)

A space is **separable** iff it has a countable dense subset, i.e. it has a countable subset s.t. every nonempty open set of X intersects the countable subset, or equivalently the closure of the countable subset is the whole space.

Lemma 5.2

Second countability is preserved under subspaces and countable products.

Definition 5.3 (Lindelöf space)

A space is **Lindelöf** iff every open cover has a countable subcover.

Lemma 5.4

Second countability implies first countability, separability, and Lindelöf.

Lemma 5.5

Any collection $\mathcal{U}$ of open subsets of a second countable space has a countable subcollection $\mathcal{V}$ s.t.

$$\bigcup_{V \in \mathcal{V}} V = \bigcup_{U \in \mathcal{U}} U$$

Thus, every open cover of a second countable space has a countable subcover, so all second countable spaces are Lindelöf.

Lemma 5.6

All regular Lindelöf spaces are normal.

In metric and pseudometric spaces, separability, Lindelöf, and second countability are equivalent.

Definition 5.7 (Separation)

Given $A \subset X$, two sets B and C are a **separation** of A iff they are nonempty and

$$B \cap \overline{C} = \emptyset = \overline{B} \cap C$$

Theorem 5.8

A subspace $A \subset X$ is disconnected iff there exists a separation of it.

6. Connectedness

See 18 Connectedness

Definition 6.1 (Connected space)

A topological space where the only sets that are both open and closed are the whole space and $\emptyset$.

Theorem 6.2

The continuous image of a connected space is connected.

Definition 6.3 (Separation)

A **separation of A** is a pair B, C of nonempty subsets of A which satisfies

$$B \cap \overline{C} = \overline{B} \cap C = \emptyset$$

Theorem 6.4

A subspace A of X is disconnected under the subspace topology iff it has a separation.

Definition 6.5 (Topological sum)

If A and B are nonempty, disjoint, and open, then $X = A \cup B$ is the **topological sum** of A and B , denotes $A \oplus B$.

Such a space is obviously disconnected.

Corollary 6.5.1

If A is connected and $A \subset B \cup C$, where $B \cap \overline{C}$ and $\overline{B} \cap C$ are both empty, though neither B nor C are, then either $A \subset B$ or $A \subset C$.

If A is a connected subset of a connected space, and $X - A = B \cup C$ where B and C are separated, then $A \cup C$ and $B \cup C$ are both connected.

If A and B are closed sets s.t. both $A \cap B$ and $A \cup B$ are connected, then A and B are both connected.

If A is connected and $A \subset B \subset \overline{A}$, then B is connected.

Let $\mathcal{A}$ be any collection of connected subsets of X with the property that any two elements of $\mathcal{A}$ have nonempty intersections. Then $\mathcal{A} = \bigcup_{A \in \mathcal{A}} A$ is connected.

X is connected iff for each pair of points, there is a connected subset of X containing both of them.

Theorem 6.6

The connected subsets of $\mathbb{R}$ are the intervals (a, b) , $(a, b]$, $[a, b)$, and $[a, b]$ where a and b are real numbers or infinite.

Theorem 6.7 (Intermediate Value Theorem)

If $f : X \rightarrow \mathbb{R}$ is continuous with X connected and $x, y \in X$, then for any $c \in (f(x), f(y))$ there exists $z \in X$ s.t. $f(z) = c$.

Theorem 6.8

A product of connected spaces is connected. Conversely, given a nonempty product that is connected, each factor is connected.

6.1. Components

Definition 6.1.1 (Component)

A connected subset of X not contained in any other connected subset of X .

Lemma 6.1.2

The union of all connected subsets of X containing x is a component containing x .

Lemma 6.1.3

The components of a space form a partition of the space.

Lemma 6.1.4

The components of a product space are the products of the components of the factors.

Definition 6.1.5 (Totally disconnected space)

A space in which each component is a single point.

Lemma 6.1.6

Products of totally disconnected spaces are totally disconnected.

Discrete spaces and the Cantor set are both totally disconnected.

6.2. Local connectedness

Definition 6.2.1 (Locally connected space)

A space in which each point has a neighborhood base consisting of connected sets.

Theorem 6.2.2

The **finite** product of locally connected sets are locally connected. If a product is locally connected, then all but finitely many factors are connected and the remainder locally connected.

Theorem 6.2.3

The following are equivalent:

1. X is locally connected.
2. If N is any neighborhood of x there is a neighborhood M of x contained in the component of N that contains x .
3. Components of open sets are open.

7. Compactness

See 19.1 Types of Compactness

Definition 7.1 (Compactness)

A space where every open cover has a finite subcover.

Definition 7.2 (Countable compactness)

A space where every countable cover has a finite subcover.

Lemma 7.3

A space is compact iff it is countably compact and Lindelöf.

Definition 7.4 (Sequential compactness)

A space where every sequence has a convergent subsequence.

Lemma 7.5

Sequential compactness implies countable compactness.

Example. Closed, bounded intervals on the real line are compact and sequentially compact.

The reals are neither compact nor sequentially compact.

$[0, \Omega]$ is both compact and sequentially compact.

$[0, \Lambda)$ for any infinite Λ is not compact, but $[0, \Omega)$ is sequentially compact.

Infinite discrete spaces are neither compact nor sequentially compact.

Definition 7.6 (Finite intersection property)

A collection of subsets of X has the *finite intersection property* iff every finite

subcollection of the sets has a nonempty intersection.

If X is a topological space, this is equivalent to being a filter base.

Theorem 7.7

The following are equivalent:

1. X is compact.
2. Every family of closed subsets of X with the finite intersection property has nonempty intersection.
3. Every ultrafilter in X has a limit.

Corollary 7.7.1 (Tychonoff Theorem)

A product $X \prod X_\alpha$ of nonempty spaces is compact iff each factor is compact.

Theorem 7.8

In a T_1 space, the following are equivalent:

1. X is countably compact
2. Every infinite open cover has a proper subcover
3. Every infinite subset has an accumulation point
4. (**Cantor Intersection Property**) Every decreasing sequence of nonempty closed subsets has nonempty intersection.

8. Summary of characterizations

8.1. Closure

Definition 2.3.1 (Closure)

Given $A \subset X$ where X is a topological space, the *closure* of A , denoted $\bar{A}$ or $\text{cl } A$, is the intersection of all closed subsets containing A .

Theorem 2.4.4 (Neighborhood characterization of closure)

Let $x \in X$, $A \subset X$. Then $x \in \bar{A}$ iff every neighborhood of x meets A .

Corollary 2.4.10.1 (Accumulation point characterization of closure)

$$\bar{A} = A \cup A'.$$

Lemma 3.2.4 (Net characterization of closure)

If A is a subset of a topological space, $x_0 \in \bar{A}$ iff there exists a net in A converging to x_0 .

Lemma 3.3.12 (Filter base characterization of closure)

$x \in \bar{A}$ iff there is a filter base in A converging to x .

8.2. Continuity

Definition 2.5.1 (Continuous function)

$f : X \rightarrow Y$ is a continuous function at $x_0 \in X$ iff $f^{-1}(N)$ is a neighborhood of x_0 for every neighborhood N of $f(x_0)$.

Equivalently, it is continuous iff every neighborhood of $f(x)$ pulls back (by f^{-1}) to a neighborhood of x .

f is continuous if it is continuous at all $x_0 \in X$.

Theorem 2.1.2 (Metric characterization of continuity)

$f : M \rightarrow N$ is continuous at $x \in M$ iff for any $\varepsilon > 0$, there exists $\delta > 0$ s.t. $d(x, y) < \delta \implies d(f(x), f(y)) < \varepsilon$.

Theorem 2.1.4 (Epsilon neighborhood definition of continuity)

$f : M \rightarrow N$ is continuous at $x \in M$ iff for every open set $V \subset N$ that contains $f(x)$, there is an open set $U \subset M$ containing x s.t. $f(U) \subset V$.

Lemma 3.2.5 (Net characterization of continuity)

$f : X \rightarrow Y$ is continuous at x_0 iff for each net $x_\alpha \rightarrow x_0$, $f(x_\alpha) \rightarrow f(x_0)$.

Lemma 3.3.11 (Filter characterization of continuity)

Let $f : X \rightarrow Y$, and $x_0 \in X$ where N is the set of neighborhoods of x_0 . Then, f is continuous at x_0 iff $N \rightarrow f(x_0)$.

9. Index

9.1. Headings

1. Sets, relations, functions, and order	1
1.1. Logic	1
1.2. Sets	1
1.3. Relations	1
1.3.1. Equivalence relations	1
1.3.2. Functions (maps)	2
1.4. Ordering	2
1.4.1. Partial and total ordering	2
1.4.2. Lattices	3
1.4.3. Order-preserving	3
1.4.4. Well-ordering	3
1.5. Cardinality	3
1.6. Countability	3
1.7. Axiom of choice	4
2. Topologies	4
2.1. Metric spaces	4
2.1.1. Epsilon neighborhoods	4
2.1.2. Subspaces	4
2.1.3. Common metrics	4

2.2. Topologies	5
2.2.1. Common topologies	5
2.3. Closure, interior, boundary	5
2.3.1. Bases	6
2.4. Neighborhoods	6
2.4.1. Neighborhood bases	7
2.4.2. Accumulation points	7
2.5. Continuity	7
2.6. Initial (induced) topology	7
2.6.1. Subspace topology	8
2.6.2. Initial (induced) topology	8
2.6.3. Product topology	8
2.6.4. Box topology	8
2.7. Covers and defining families	9
2.8. Final (co-induced) topologies	9
2.8.1. Quotient topology	9
3. Limits	9
3.1. Sequences	9
3.2. Nets	9
3.3. Filters	10
3.3.1. Ultrafilters	10
4. Separation	11
4.1. Separation properties	11
4.2. T_0 through T_2 spaces	11
4.3. Regular spaces	11
4.4. Normal spaces	12
5. Countability properties	12
6. Connectedness	13
6.1. Components	13
6.2. Local connectedness	13
7. Compactness	14
8. Summary of characterizations	14
8.1. Closure	14
8.2. Continuity	14
9. Index	15
9.1. Headings	15
9.2. Definitions	15
9.3. Theorems	16
9.4. Axioms	17
9.5. Lemmata	17

9.6. Corollary	17
----------------	----

9.2. Definitions

Definition 1.2.1	Power set	1
Definition 1.3.1	Relation	1
Definition 1.3.4		1
Definition 1.3.5	Equivalence relation	1
Definition 1.3.6	Equivalence class	1
Definition 1.3.8	Partition	1
Definition 1.3.10	Function	2
Definition 1.3.11	Injective	2
Definition 1.3.12	Surjective	2
Definition 1.3.13	Bijjective	2
Definition 1.3.15		2
Definition 1.3.18	Induced map	2
Definition 1.4.1	Partial order	2
Definition 1.4.2	Total order	2
Definition 1.4.3	Strict order	2
Definition 1.4.4	Partially ordered and totally ordered set	2
Definition 1.4.5	Upper and lower bounds	2
Definition 1.4.6	Lattice	3
Definition 1.4.8	Order-preserving map	3
Definition 1.4.10	Order isomorphism	3
Definition 1.4.11	Well-ordered set	3
Definition 1.4.12	Initial segment	3
Definition 1.5.1	Cardinality	3
Definition 1.5.2		3
Definition 1.6.1	Countability	3
Definition 2.1.1	Metric	4
Definition 2.1.3	Epsilon neighborhood	4
Definition 2.1.5	Subspace of a metric space	4
Definition 2.1.6	Standard metric on $\mathbb{R}^n$	4
Definition 2.1.7	Taxicab metric on $\mathbb{R}^n$	4
Definition 2.1.8	Chebyshev metric on $\mathbb{R}^n$	5
Definition 2.1.9	Paris metro metric	5
Definition 2.1.10	Discrete metric	5
Definition 2.1.11	Indiscrete pseudometric	5
Definition 2.2.1	Topology	5
Definition 2.2.2	Closed set	5

Definition 2.2.4	Metrizable topology	5	Definition 3.2.3	Convergence of a net	9	Definition 7.6	Finite intersection property . . .	14
Definition 2.2.5	Finer, coarser topologies	5	Definition 3.3.1	Filter base	10	9.3. Theorems		
Definition 2.2.7	Standard topology on $\mathbb{R}$	5	Definition 3.3.2	Filter	10	Theorem 1.1.1	De Morgan's Laws for statements	1
Definition 2.2.8	Discrete topology	5	Definition 3.3.4	Canonical filter	10	Theorem 1.1.2	De Morgan's Laws for set theory	1
Definition 2.2.9	Indiscrete topology	5	Definition 3.3.5	Fréchet filter	10	Theorem 1.1.3		1
Definition 2.2.10	Metric topology	5	Definition 3.3.7	Elementary filter	10	Theorem 1.3.2	Relations can compose	1
Definition 2.2.11	Cofinite topology	5	Definition 3.3.8	Convergence of a filter base . .	10	Theorem 1.3.3		1
Definition 2.2.12	Sorgenfrey line	5	Definition 3.3.15	Ultrafilter	10	Theorem 1.3.9		2
Definition 2.3.1	Closure	5	Definition 3.3.17	Cluster point	10	Theorem 1.3.14		2
Definition 2.3.3	Interior	5	Definition 4.1.1	Separable by neighborhoods . .	11	Theorem 1.3.17		2
Definition 2.3.4	Boundary	5	Definition 4.1.2	Separation by closed neighborhoods	11	Theorem 1.4.7		3
Definition 2.3.5	Exterior	6	Definition 4.1.3	Separation by continuous functions	11	Theorem 1.4.9		3
Definition 2.3.7	Base	6	Definition 4.2.1	Kolmogorov space (T_0)	11	Theorem 1.4.14		3
Definition 2.3.8	Second countable	6	Definition 4.2.3	Fréchet space (T_1)	11	Theorem 1.5.3	Schroeder-Bernstein Theorem . .	3
Definition 2.3.10	Sub-base	6	Definition 4.2.6	Hausdorff space (T_2)	11	Theorem 1.5.4	Cantor's Theorem	3
Definition 2.3.11	Topology generated by a sub- base	6	Definition 4.2.7	$T_{2\frac{1}{2}}$ space	11	Theorem 1.6.2		3
Definition 2.4.1	Neighborhood	6	Definition 4.2.8	Functionally Hausdorff space .	11	Theorem 1.6.3		4
Definition 2.4.6	Neighborhood base	7	Definition 4.3.1	Regular space	11	Theorem 1.7.2	Zermelo's Theorem	4
Definition 2.4.7	First countable	7	Definition 4.3.3	Regular Hausdorff space (T_3) .	11	Theorem 1.7.3	Well-Ordering Property (Zermelo's Postulate)	4
Definition 2.4.9	Accumulation point	7	Definition 4.3.4	Completely regular space	11	Theorem 1.7.4	Hausdorff Maximum Principle . .	4
Definition 2.4.10	Derived set	7	Definition 4.3.5	Tychonoff space ($T_{3\frac{1}{2}}$)	11	Theorem 1.7.5	Zorn's Lemma	4
Definition 2.5.1	Continuous function	7	Definition 4.4.1	Normal space	12	Theorem 2.1.2	Metric characterization of continuity	4
Definition 2.5.2	Homeomorphism	7	Definition 4.4.2	Normal Tychonoff space (T_4) .	12	Theorem 2.1.4	Epsilon neighborhood definition of continuity	4
Definition 2.6.1	Subspace topology	8	Definition 4.4.4	Completely normal	12	Theorem 2.3.2	Kuratowski closure axioms	5
Definition 2.6.3	Initial topology	8	Definition 4.4.5	Completely normal Tychonoff (T_5)	12	Theorem 2.3.6		6
Definition 2.6.6	Open and closed functions	8	Definition 5.1	Separable	12	Theorem 2.3.12	Topology generated by a base . .	6
Definition 2.6.7	Projection	8	Definition 5.3	Lindelöf space	12	Theorem 2.4.4	Neighborhood characterization of closure	6
Definition 2.6.8	Product topology	8	Definition 5.7	Separation	12	Theorem 2.4.5	Neighborhood characterization of boundary	6
Definition 2.6.11	Box topology	8	Definition 6.1	Connected space	13	Theorem 2.5.3	Continuity composes	7
Definition 2.7.1	Defining family	9	Definition 6.3	Separation	13	Theorem 2.5.4		7
Definition 2.7.2	Open cover	9	Definition 6.5	Topological sum	13	Theorem 2.6.5		8
Definition 2.7.3	Locally finite	9	Definition 6.1.1	Component	13	Theorem 2.6.10		8
Definition 2.7.4	Closed cover	9	Definition 6.1.5	Totally disconnected space	13	Theorem 2.7.5		9
Definition 2.8.1	Final topology	9	Definition 6.2.1	Locally connected space	13	Theorem 3.1.5		9
Definition 2.8.2	Quotient topology	9	Definition 7.1	Compactness	14			
Definition 3.1.1	Sequence	9	Definition 7.2	Countable compactness	14			
Definition 3.1.2	Convergence of a sequence	9	Definition 7.4	Sequential compactness	14			
Definition 3.2.1	Directed upward	9						
Definition 3.2.2	Net	9						

Theorem 3.1.6	9
Theorem 3.3.13	10
Theorem 3.3.16 Test for ultrafilter	10
Theorem 3.3.18	10
Theorem 4.3.2	11
Theorem 4.3.6	11
Theorem 4.3.7	12
Theorem 4.4.6 Urysohn’s Lemma	12
Theorem 4.4.7 Tietze Extension Theorem	12
Theorem 5.8	12
Theorem 6.2	13
Theorem 6.4	13
Theorem 6.6	13
Theorem 6.7 Intermediate Value Theorem	13
Theorem 6.8	13
Theorem 6.2.2	13
Theorem 6.2.3	13
Theorem 7.7	14
Theorem 7.8	14

9.4. Axioms

Axiom 1.7.1 Axiom of Choice	4
Axiom 2.4.3 Hausdorff neighborhood axioms	6
Axiom 2.4.8 Neighborhood axioms	7

9.5. Lemmata

Lemma 1.3.7	1
Lemma 1.3.16	2
Lemma 1.4.13	3
Lemma 2.2.3	5
Lemma 2.2.6	5
Lemma 2.3.9	6
Lemma 2.4.2	6
Lemma 2.5.5	7
Lemma 2.5.6	7
Lemma 2.6.2	8
Lemma 2.6.4	8
Lemma 2.6.9 Product topology constrains finitely many axes	8
Lemma 3.1.3	9

Lemma 3.1.4	9
Lemma 3.2.4 Net characterization of closure	10
Lemma 3.2.5 Net characterization of continuity	10
Lemma 3.3.3	10
Lemma 3.3.6 Images of filter bases are filter bases	10
Lemma 3.3.9 Constructing filter base from net	10
Lemma 3.3.10 Collection of neighborhoods is filter	10
Lemma 3.3.11 Filter characterization of continuity	10
Lemma 3.3.12 Filter base characterization of closure	10
Lemma 3.3.14	10
Lemma 4.2.2	11
Lemma 4.2.4	11
Lemma 4.2.5	11
Lemma 4.2.9	11
Lemma 4.2.10	11
Lemma 4.4.3	12
Lemma 5.2	12
Lemma 5.4	12
Lemma 5.5	12
Lemma 5.6	12
Lemma 6.1.2	13
Lemma 6.1.3	13
Lemma 6.1.4	13
Lemma 6.1.6	13
Lemma 7.3	14
Lemma 7.5	14

9.6. Corollary

Corollary 1.4.9.1	3
Corollary 1.6.1.1	3
Corollary 2.4.10.1 Accumulation point characterization of closure	7
Corollary 4.3.7.1	12
Corollary 4.4.7.1	12
Corollary 6.5.1	13
Corollary 7.7.1 Tychonoff Theorem	14